DATA ANALYSIS PROJECT · DIGITERA TRAINING

PharmaPulse Analytics

Sales, Loss & Inventory Intelligence for Pharmacy Chains

End to end: Python · SQL Server · Power BI · DAX

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The business problem

A pharmacy chain with 12 stores across 5 regions wants to know where profit leaks, which discounts pay off, and where stock is at risk.

Raw data	Python + Power Query	SQL Server	Power BI	Insights
3 sales files + 8 tables	Clean and merge	Explore and validate	Model, DAX, dashboard	Decisions

THE QUESTIONS BEHIND THE PROJECT

- **Where does revenue and profit come from?**
- **Which sales lose money, and why?**
- **Which discounts and promo codes are worth it?**
- **Where is inventory at risk: low stock, items close to expiry?**

Python: cleaning the data

```
# two date formats in the raw files
is_slash = d.str.match(r"^\d{2}/\d{2}/\d{4}$")
parsed[is_slash]  = to_datetime(d[is_slash],  format="%d/%m/%Y")
parsed[~is_slash] = to_datetime(d[~is_slash], format="%Y-%m-%d")
# flag the sales that lose money
sales["is_loss"] = (sales["profit"] < 0).astype(int)
```

What the script handles

- Merges 3 sales files with different column names and two date formats
- Removes 50 duplicate rows and 42 duplicate customers
- Extracts promo codes and discount reasons from free-text notes
- Adds profit, margin, is_loss, discount_band

POWER QUERY (EXCEL)

Cleaned and shaped the other 8 tables: products, customers, stock, prescriptions, employees, stores, suppliers and categories.

WHAT IT GAVE US

A trusted base: 9,950 rows became 9,900 after de-duplication, with 0 failed date conversions. The is_loss flag made the loss analysis possible: 608 loss sales (6.14%).

SQL: do bigger discounts hurt margin?

```
SELECT discount_band,
  ROUND(100.0*SUM(profit)/SUM(total_amount),2) AS margin_pct,
  ROUND(100.0*SUM(is_loss)/COUNT(*),2) AS loss_rate_pct
FROM dbo.Sales_cleaned
GROUP BY discount_band;
```

Discount	Margin	Loss rate
0-5%	40.9%	4.8%
5-10%	38.7%	5.2%
10-15%	36.5%	5.2%
15%+	21.3%	9.2%

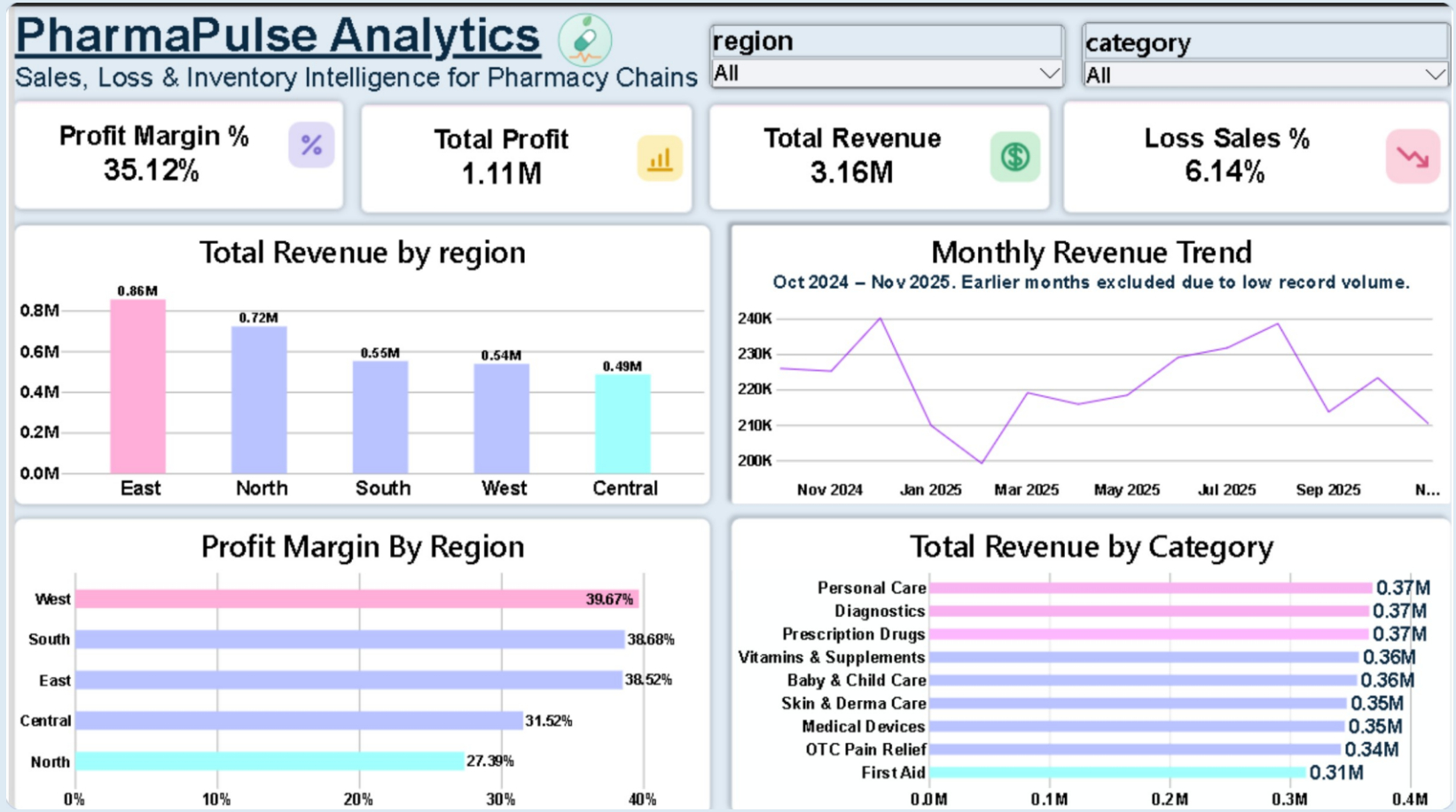
SQL SERVER

22 queries: JOINs, GROUP BY, HAVING, CTEs and window functions (LAG, RANK). Also used to cross-check the dashboard totals: Revenue 3.16M, Profit 1.11M, 608 loss sales.

WHAT IT REVEALED

Discounts above 15% destroy margin: 21.3% vs 40.9% at 0-5%, and the loss rate doubles to 9.2%. These sales are about 25% of all sales.

Overview: East earns the most, West has the best margin



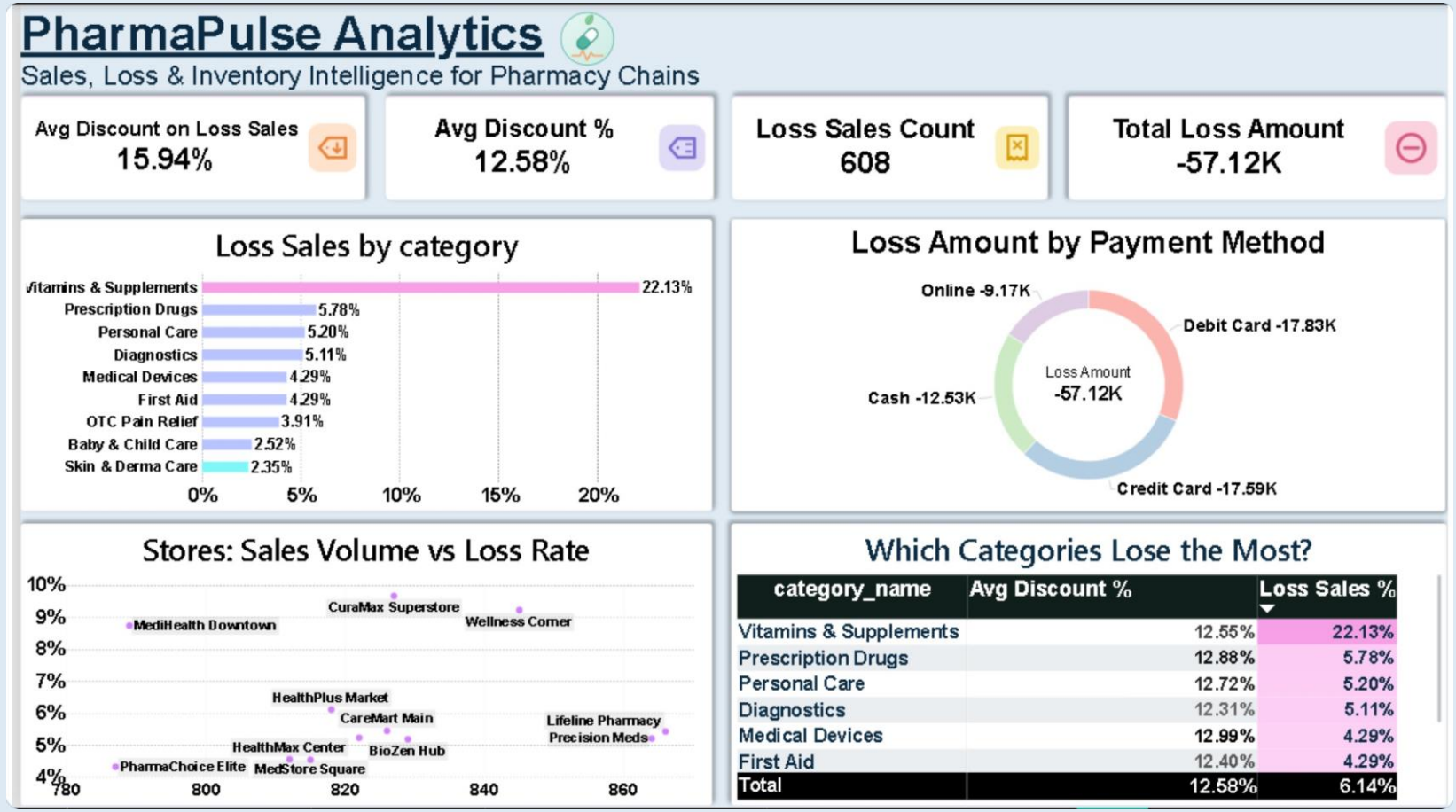
INSIGHTS

- East earns the most revenue (0.86M). West has the highest margin (39.7%).
- North is the 2nd highest region by revenue (0.72M) but has the lowest margin (27.4%).
- Next step: investigate North's pricing and discount practice.

POWER BI · DAX

Star schema model with DAX KPIs (Profit Margin %, Loss Sales %, ...). Overview page: Revenue 3.16M, Profit 1.11M, margin 35.12%.

Loss: Vitamins lose 22% of their sales



INSIGHTS

- Vitamins & Supplements has a 22.1% loss rate, against 2-6% for every other category.
- It is a pricing problem, not a discount problem: it loses 19-21% of sales even at 0-10% discounts.
- 112 of its 543 products (21%) have a cost price above the selling price, vs about 2-4% in other categories.

POWER BI · DAX

Built on the is_loss flag from Python: 608 loss sales worth -57.12K, sliced by category, payment method and store.

Inventory: 31% of stock records are below reorder level



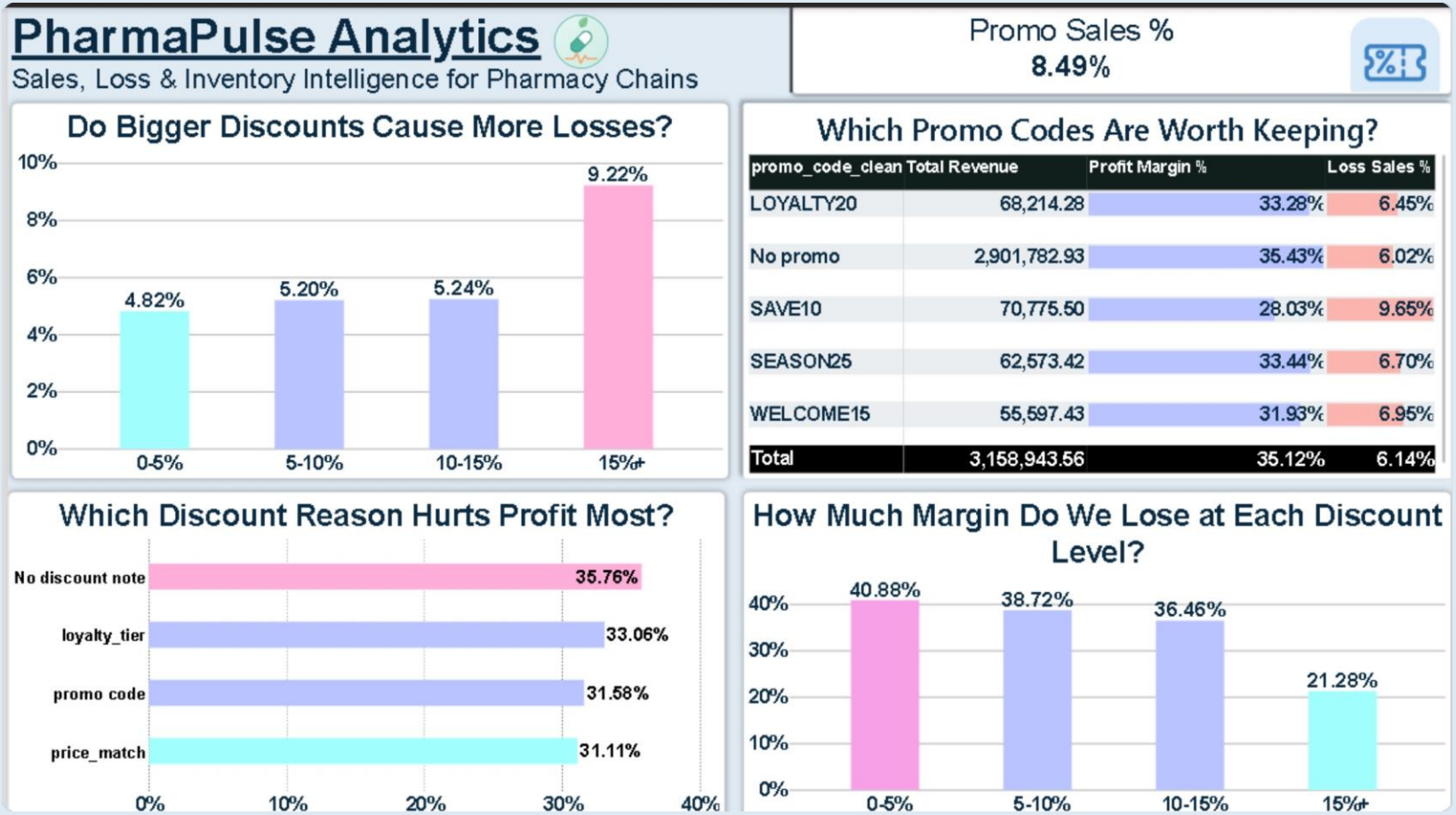
INSIGHTS

- About 31% of stock records are below reorder level (3.1K of 10K).
- Expiring items peak in September 2026 (355 items).
- Next step: set a clear "expiring soon" window and clear or return near-expiry stock before the monthly peaks.

POWER BI · DAX

Inventory page: low stock by store and by supplier, expiring-soon items by month, and a gauge of shortage vs target.

Discounts: above 15%, the margin is cut in half



INSIGHTS

- Above 15% the margin falls to 21.3% (vs 40.9% at 0-5%) and the loss rate rises to 9.2%.
- Promo code sales earn a lower margin (31.6%) than regular sales (35.4%).
- SAVE10 is the weakest code: 28.0% margin and 9.7% loss rate.

POWER BI · DAX

Discounts page: margin and loss rate by discount band, promo code and discount reason. Promo sales are 8.49% of all sales.

What the data says

21.3%

margin on discounts above 15%, against 40.9% for 0-5%. Loss rate doubles to 9.2%.

22.1%

loss rate in Vitamins. 112 of its 543 products are priced below cost.

28.0%

margin on promo code SAVE10, the weakest code, with a 9.7% loss rate.

RECOMMENDATIONS

1. Cap discounts at around 15% and require approval above it.
2. Fix the selling price (or supplier cost) of the Vitamins products priced below cost.
3. Review or retire the SAVE10 promo code.
4. Investigate North's pricing and discount practice.
5. Set a clear "expiring soon" window and clear or return near-expiry stock before the peaks.

Thank you

Python · SQL Server · Power BI · DAX · Excel

Full code, queries and dashboard are on GitHub

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